

Disproof of Conjecture 00000000156

Prime determinants of random sign matrices

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Verdict: false. For every $n \geq 3$, the probability in the conjecture is exactly zero. The general matrix obstruction, the probability statement, and the failure of the proposed asymptotic are all verified in Lean 4.

1 Statement and probability model

The conjecture asserts that the probability that the determinant of a random $n \times n$ matrix with entries in $\{1, -1\}$ is prime in absolute value is asymptotic to c/n , with an explicit constant c . We use the usual uniform model: the n^2 signs are independent and equally likely. Let

$$p_n = \frac{\#\{A \in \{1, -1\}^{n \times n} : |\det A| \text{ is prime}\}}{2^{n^2}}.$$

Here primes are positive integers at least 2. The standard meaning of $p_n \sim c/n$ for $c > 0$ is $\lim_{n \rightarrow \infty} p_n/(c/n) = 1$.

2 A uniform obstruction

Theorem 1. *For every integer $n \geq 3$ and every sign matrix A of size n , $4 \mid \det A$. Consequently, $|\det A|$ is not prime.*

Proof. Subtract row 0 from rows 1 and 2. These two row operations preserve the determinant. Each entry in either resulting row is the difference of two signs, so it is in $\{-2, 0, 2\}$. Define integral row vectors

$$u_j = (A_{1j} - A_{0j})/2, \quad v_j = (A_{2j} - A_{0j})/2.$$

The transformed rows are $2u$ and $2v$. By multilinearity in the rows,

$$\det A = 4 \det C,$$

where C is the integral matrix obtained by replacing rows 1 and 2 of A with u and v . Thus $4 \mid \det A$ and $4 \mid |\det A|$. If $|\det A|$ were prime, its divisor 2 would force $|\det A| = 2$; but 4 does not divide 2. This is a contradiction, including the case $\det A = 0$, since zero is not prime. $\square$

Subtracting row 0 from *all* other rows also gives the stronger familiar fact $2^{n-1} \mid \det A$. The Lean proof uses just two rows, since divisibility by 4 already proves the required obstruction in every dimension $n \geq 3$.

3 Probability and asymptotics

The theorem makes the event defining p_n empty for every $n \geq 3$. Therefore $p_n = 0$ for all $n \geq 3$. For every $c > 0$, the ratio $p_n/(c/n)$ is eventually identically zero, hence converges to zero and cannot converge to one. This disproves the conjecture. Choosing $c = 0$ cannot rescue a standard asymptotic equivalence: its comparison function is zero and the usual ratio definition is undefined. In Lean, real division is total; even with that convention, the ratio limit is zero for every real c .

For context, $p_1 = 0$ and $p_2 = 1/2$. Indeed, for $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $\det A = ad - bc$, and the two independent uniform signs ad and bc differ with probability $1/2$. These small cases are not needed for the disproof.

4 Formal verification and previous submission

The directory `lean/` contains a pinned Mathlib-based Lean 4 project. `Main.lean` proves divisibility by 4 for arbitrary $n \geq 3$ and arbitrary sign matrices, then proves the non-primality of the absolute determinant. Boolean matrices encode the uniform finite sample space; `primeProbability` is its favorable count divided by its total count. The final theorem `conjecture_false` rules out any positive c with the stated ratio limit. No finite enumeration substitutes for a general theorem.

The earlier merged PR #113 was subsequently removed by semantic review in commit 4504549. Its written mathematical argument was correct, but its Lean verification covered only conditional arithmetic and finite examples rather than the general matrix and asymptotic statements. This submission supplies those missing formal results.

Reproduction, from `lean/`:

```
lake update
lake exe cache get
lake build
lake env lean Check.lean
```

The axiom audit uses only Lean/Mathlib's standard foundations (`propext`, `Classical.choice`, `Quot.sound`); there are no admitted proofs, additional axioms, or `native_decide`.